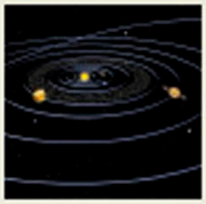
# Space

The Universe is thought to have been created about 14 billion years ago in a colossal explosion known as the Big Bang. Since then it has been evolving and expanding at an ever-increasing rate. For centuries, astronomers were only able to study the celestial bodies they could observe from Earth. Today, they can explore the Universe at large using a wide range of sophisticated instruments, from powerful telescopes to robotic spacecraft.

## What's in space?

The most common objects in the Universe are the billions of stars that form the galaxies. Our local star, the Sun, is just one of the millions of stars in a galaxy called the Milky Way. The Sun and the planets that orbit it make up the solar system, together with other celestial bodies such as moons, comets, and asteroids.

**Galaxies**
There are more than a hundred billion galaxies in the Universe. Each one consists of a vast collection of stars, gas, and dust, held together by gravity.



**Planets**
Planets are balls of matter that orbit the Sun. They were created more than 4.6 billion years ago from gas and dust left over when the Sun was formed. Most other stars also have planets.

**Nebulas**
Named after the Latin word for "cloud," nebulas (singular nebula) are massive clouds of gas and dust. All the stars in the Universe are created from nebulas.

**Moons**
A moon is a small, celestial body that orbits a planet. All the planets in our solar system have moons except for Mercury and Venus.

**Stars**
There are more stars than any other object in the Universe. Each is a spinning ball of hot, luminous gas. Most stars are made entirely of hydrogen and helium.

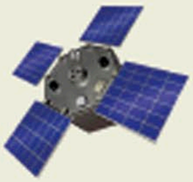
**Other celestial bodies**
Smaller bodies in the solar system include comets, which are frozen balls of gas and dust, and asteroids, which are pieces of rock and metal that orbit the Sun.

## Space exploration

The space age began in 1957, when the first satellite was launched. Since then, hundreds of astronauts and robotic spacecraft have traveled from Earth to explore the Universe. Robotic (unmanned) space probes are cheaper and safer than manned spacecraft as they can travel in space without the need for astronauts.

**Telescopes**
Space telescopes collect light and other forms of energy from stars and galaxies to give a better view of the Universe.

**Spacecraft**
Robotic spacecraft have been used since 1959 to make long-distance journeys to the planets and their moons.



**Satellites**
Satellites orbiting Earth have many uses, such as relaying telephone calls or providing data for weather forecasts.

**Space stations**
Space stations provide a base for astronauts to live and work in, and serve as a launch pad for space missions.

**Rockets**
A rocket is used to send a satellite or spacecraft into space. The rocket's cargo of equipment or crew is called the payload.